

1) Find 10 different model providers and their models released in last 12 months

Model Provider	Model	Release Date	Cut-off Date	Cost of Tokens	Context Window	Type	When to Use
Model Provider	Model	Release Date	Cut-off Date	Cost of Tokens	Context Window	Type	When to Use
OpenAI	GPT-5.6 Sol	7/9/2026	Early 2026	~\$5 per 1M tokens	2M tokens	General reasoning	Enterprise-grade reasoning, long-context workflows
OpenAI	GPT-5.6 Terra	7/9/2026	Early 2026	~\$3 per 1M tokens	2M tokens	General reasoning	Automation, RAG pipelines
OpenAI	GPT-5.6 Luna	7/9/2026	Early 2026	~\$1.5 per 1M tokens	2M tokens	General reasoning	Cost-efficient reasoning tasks
OpenAI	GPT-5.5	2026	Late 2025	~\$1 per 1M tokens	1M tokens	General-purpose LLM	Coding, structured generation
Anthropic	Claude Opus 5	7/24/2026	Early 2026	~\$15 per 1M tokens	1M tokens	High-end reasoning	Deep analysis, legal reasoning
Anthropic	Claude Sonnet 5	6/30/2026	Early 2026	~\$3 per 1M tokens	1M tokens	Mid-tier reasoning	Fast, cost-efficient reasoning
Google DeepMind	Gemini 3.5 Flash Cyber	7/21/2026	Early 2026	~\$0.30 per 1M tokens	2M tokens	Cybersecurity-tuned	Threat detection, secure code review
Google DeepMind	Gemini 3.5 Flash	2026-07	Early 2026	~\$0.20 per 1M tokens	2M tokens	Lightweight model	High-volume tasks, chatbots
xAI	Grok 4.5	7/8/2026	Early 2026	~\$1 per 1M tokens	1M tokens	General-purpose LLM	Real-time tasks, conversational AI
Zhipu AI	GLM-5.2	6/13/2026	Early 2026	~\$0.50 per 1M tokens	1M tokens	Multilingual LLM	Chinese/English bilingual tasks
Moonshot AI	Kimi K3	7/16/2026	Early 2026	~\$0.40 per 1M tokens	2M tokens	Long-context reasoning	Research, document-heavy workflows
Moonshot AI	Kimi K2.7 Code	6/12/2026	Early 2026	~\$0.25 per 1M tokens	1M tokens	Code-specialized	Code generation, debugging

DeepSeek	V4 Flash (0731)	7/31/2026	Early 2026	~\$0.14 per 1M tokens	1M tokens	High-speed inference	Large-scale automation
Alibaba	Qwen3.8-Max Reasoning	8/3/2026	Early 2026	~\$0.60 per 1M tokens ²	2M tokens	Reasoning-optimized	Enterprise logic tasks
Alibaba	Qwen3.7-Flash	7/27/2026	Early 2026	~\$0.20 per 1M tokens	1M tokens	Lightweight model	Rapid generation, automation
ByteDance	Doubao Seed 2.1 Pro	6/24/2026	Early 2026	~\$0.15 per 1M tokens	1M tokens	General-purpose	High-volume consumer apps
ByteDance	Doubao Seed 2.1 Turbo	6/24/2026	Early 2026	~\$0.10 per 1M tokens	1M tokens	General-purpose	Multilingual tasks
Meituan	LongCat-2.0	6/29/2026	Early 2026	~\$0.25 per 1M tokens	2M tokens	Long-context model	Summarization, RAG

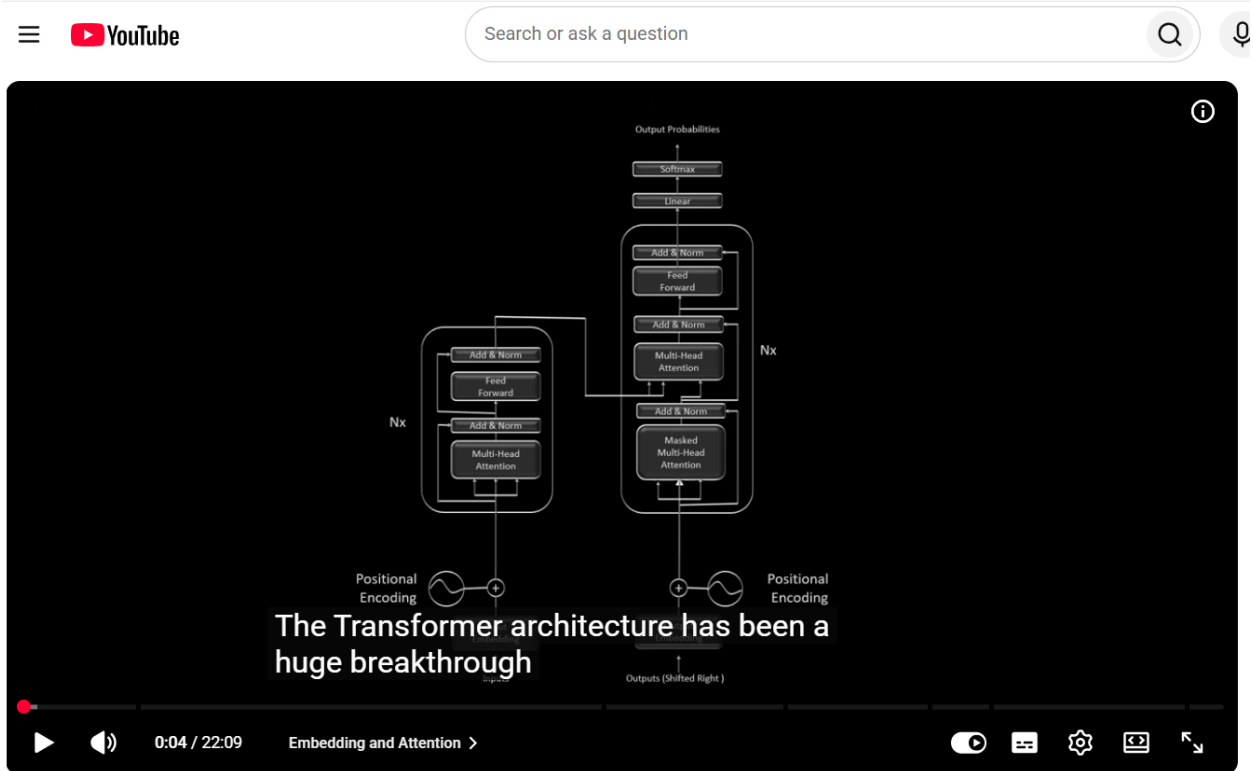
2. Read 2–3 articles about how transformer works?

Goal to understand:

- (a) Self-attention mechanism
- (b) Vector embedding
- (c) Feed forward Neural Network

Article #1

Article: "How Attention Works in Transformers | AP Lab Academy"



Article #2

Article: "Transformers from First Principles: Part 2 - Embeddings | Matías Battaglia"

The presentation slide illustrates the Transformer architecture. On the left, the sentence "I made kheer" is shown above a matrix of contextual embeddings. The middle section features a detailed diagram of the Transformer encoder block, which includes layers for Multi-Head Attention, Feed Forward, and Positional Encoding, all connected by residual connections and layer normalization. On the right, a list of words in Hindi is shown with their corresponding probabilities: आम (0.9%), कौआ (2.3%), खीर (27.9%), सूरज (2.4%), and ज्ञान (0.07%). A video thumbnail of a man speaking is located in the bottom right corner.

previous step you feed that okay as an input here and then it will produce the

Transformers Explained | Simple Explanation of Transformers